MEITREX - Gamified and Adaptive Intelligent Tutoring in Software Engineering Education

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ABSTRACT

Nowadays, learning management systems (LMSs) are established tools in higher education, especially in the domain of software engineering (SE). However, the potential of such educational technologies has not been fully exploited, as student performance in SE education is still strongly dependent on feedback from time-constrained lecturers and tutors. Moreover, current LMSs are not designed for SE courses, as external SE tools are required to fulfill the requirements of lecturers such as programming and UML modeling features. Evolving these LMSs in the direction of intelligent tutoring could assist students in receiving automatic, individual feedback from the LMSs on their learning performance at any time. Also, gamified learning elements can serve to motivate students to engage with SE materials. Therefore, this paper presents an approach combining learning analytics, feedback, and interactive learning such as gamification in one LMS designed for SE education. The system could thus address diverse students with different backgrounds and motivational aspects and provide appropriate individual support to ensure effective SE education.

CCS CONCEPTS

• Software and its engineering; **• Social and professional topics → Computing education**; **Student assessment**; **• Information systems** → *Open source software*;

KEYWORDS

Software Engineering Education, Student Motivation, Intelligent Tutoring System, Learning Analytics, Gamification, Feedback

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1 PROBLEM

Software engineering (SE) education plays a vital role in training future software engineers and computer scientists [24]. Thus, in this research, we also explicitly include programming education. In SE education, particular emphasis is placed on developing students' practical skills [10], which requires much feedback from lecturers. However, established learning management systems (LMSs) are not designed for this practice-oriented SE education, so lecturers must rely on additional external SE tools. Utilizing blended learning [43] and remote SE education [5, 25] reduces the direct interaction between students and lecturers, resulting in less feedback on student performance. This also affects the motivation of students [2, 12, 16, 20], who have to muster more intrinsic motivation [4] to engage with SE content and keep up with the courses. Unfortunately, established LMSs barely offer lecturers any possibilities for adapting learning content to students [30] and tailoring it to the different motivational aspects of individual students [6]. In higher education, technical, teaching and learning, as well as organizational challenges complicate the realization of adaptive learning [30]. This leads to the main research question of this PhD: *"How can the motivation of students with different backgrounds and learning preferences be improved in software engineering education?"*

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2 RELATED WORK

There is research and a few developed tools in SE education that tackle similar problems. Krusche and Seitz's *ArTEMiS* [22] is an automatic assessment tool for interactive programming courses, which can also be used as an LMS and is continually being further developed. However, the individual feedback focuses on assessing students' source code and not on broader learning analytics of students in SE education, nor does it focus on adaptive learning and student motivation. There are also approaches by Striewe et al. for programming education to provide assessments and feedback for students based on their developed source code [38] in their educational system *JACK* [14] with similar drawbacks. Commercial tools such as *MATHia* by Carnegie Learning Inc. [9] are tools that analyze students based on their given answers, provide appropriate feedback and adaptive learning but are not tailored to SE education and are limited to teaching in (high-)schools. The systematic reviews by Crow et al. [11] and Nesbit et al. [33] on intelligent tutoring systems (ITSs) in computer science, software engineering, and programming education show that some researchers have developed ITSs that include gaming and allow students to program robots or similar. Another trend in some ITSs is the tracking and modeling of the emotional state of students to provide pedagogical and strategic support [33]. However, most of the ITSs described relate to programming and do not focus on adaptive learning and student motivation in SE education. Nesbit et al. also recommend in their systematic review the development of a standardized, open-source ITS for teaching introductory programming [33].

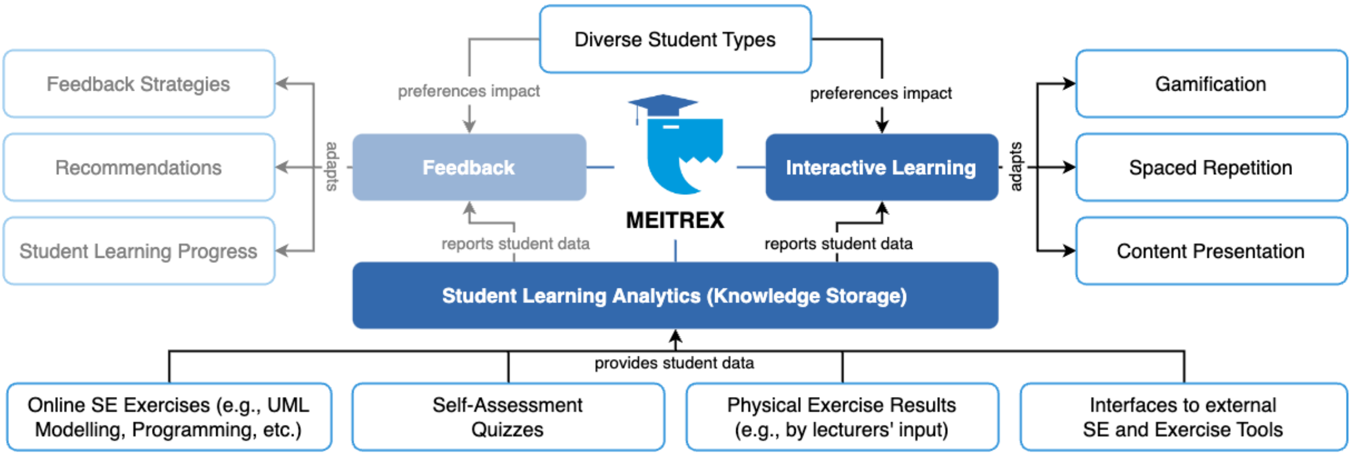

Figure 1: Overview of our approach for gamified and adaptive intelligent tutoring in SE education.

3 PROPOSED APPROACH

Addressing the identified problem, we propose the development of *MEITREX (**M**odular **E**mbedded **I**ntelligent **T**utoring and **R**emote **E**ducation e**X**perience)*, a novel approach making SE education more engaging and motivating by providing each student with individualized feedback and learning elements adapted to their learning progress. *MEITREX* is an intelligent tutoring system, tailored to higher SE education. In contrast to existing ITSs, our objective in using the system in SE education is to address student diversity and support all students independently according to their needs [13, 42]. Lecturers should not have any extra effort when using *MEITREX*. In general, ITSs assist students by assessing their individual learning progress and providing appropriate feedback that supports the learning process [15, 17, 29, 35, 41]. The integration of gamified elements can also increase students' motivation to actively engage with course material [1, 3, 18, 19, 31, 34]. *MEITREX* combines concepts from the areas of *learning analytics*, *feedback* as well as *interactive learning* such as gamification in one system and tailors them to the needs of SE education. An overview of the components of *MEITREX* is presented in Figure 1. One key aspect of the proposed system is its adaptability to meet the individual learning progress and preferences of students with different backgrounds and support them individually as effectively as possible [40, 42]. The system collects and analyzes the information of individual students (*learning analytics*), including their answers and learning behavior, preferred learning styles, learning pace, and prior knowledge. Based on the data gathered, *MEITREX* adapts its *feedback* to the individual students, such as feedback strategies [32] and recommendations, as well as the *interactive learning* such as gamification and spaced repetition [39] accordingly. The focus on students' individual learning progress enables *MEITREX* to create a personalized SE education experience. *MEITREX* will be used to test the hypothesis that a gamified and adaptive intelligent tutoring approach positively impacts student motivation and learning experience in SE education. An essential role in integrating the different concepts into one system may be *Bloom's Taxonomy* [21], where students could be categorized according to their competencies.

4 EXPECTED CONTRIBUTIONS

We will have three main contributions to support and motivate students in SE education: (1) The first contribution will be an exploration of student diversity in the SE domain and the related issues in established LMSs. (2) The second contribution will focus on student learning analytics, using a machine learning model tailored for SE education to analyze student performance. (3) The third contribution will focus on adapting interactive learning components, such as gamified elements based on individual students' preferences and performance from (1) and (2). *MEITREX's* feedback components (see Figure 1) will be contributed in parallel by other PhDs.

5 CURRENT STATUS

The PhD is currently in an early stage. Through student projects during the master's program, some ideas, visions, and considerations, which we have published, were approached but do not contribute to the PhD as they represent abstract thoughts of the domain instead of a focused plan for dissertation contributions [26, 28, 36]. Currently, we are developing the base framework for *MEITREX* to integrate learning analytics, gamification, and feedback components in the future [27]. In addition, further research has been carried out into using large language models for SE education, which also does not contribute to the PhD [37]. Still, it may be useful for a later phase of research to automatically generate SE exercises within *MEITREX*.

6 PLAN FOR EVALUATION AND VALIDATION

We plan to evaluate our approach and its expected benefits based on a Goal-Question-Metric [8] approach with a user study. As described with the research question in Section 1, the evaluation focuses on increasing student motivation and engagement in SE courses. Therefore, information on the motivation and engagement of students, as well as data on participation and examination in various SE courses at different universities, will be collected over several semesters and ultimately compared with the results achieved after the use of *MEITREX* in these courses [23]. In periodic status surveys, students of these courses are asked about their motivation, learning progress, course satisfaction, comfort, and stress levels [7].

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